

Unsupervised Representation Learning

AI 4 Life Science Summer School - Aug 19

Jeremy Tan - AAU

Motivation

- Big data
- Labelling is expensive
- Beyond labels

What is a label?

- Reference from which we can compute an error
- Consider:
 - What information is available?
 - What do we want to learn?

What we have?

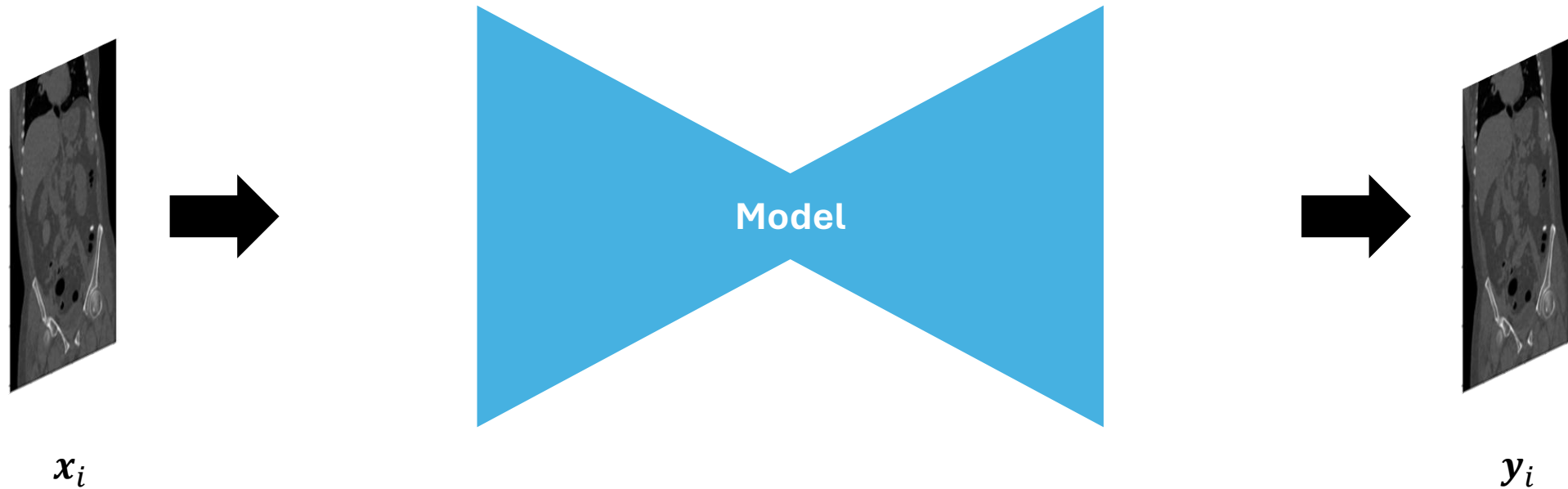
- The data itself (images)
- Domain knowledge of:
 - The sampled distribution
 - The subject
- Assumptions

Vs.

What we want?

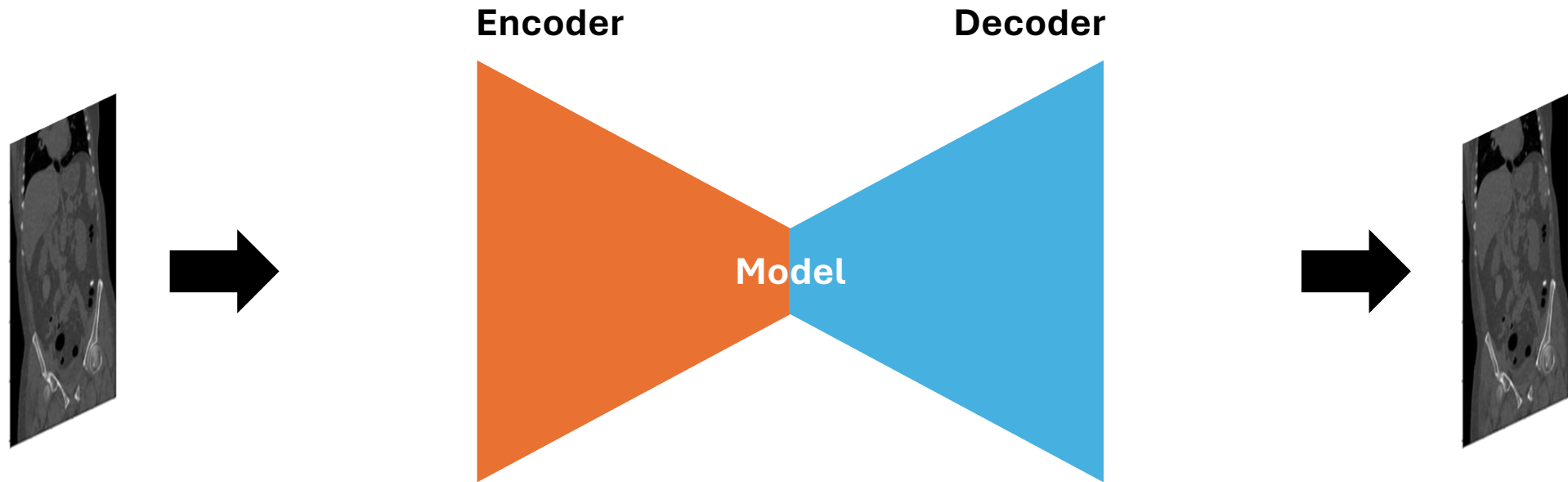
- Approximate features of supervised model
- General features for any task
- Manifold where data lies
- Cluster/biomarker discovery

Autoencoders

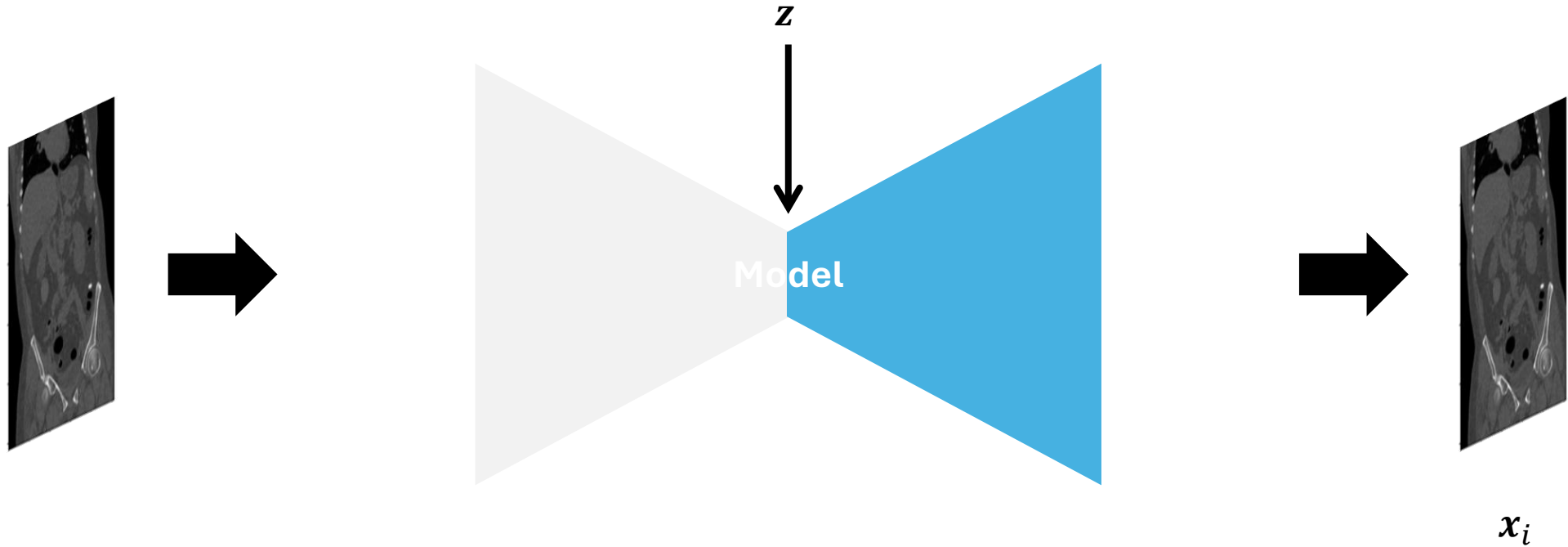


$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{y}(x_i, \mathbf{w}) - x_i\|^2$$

Autoencoders



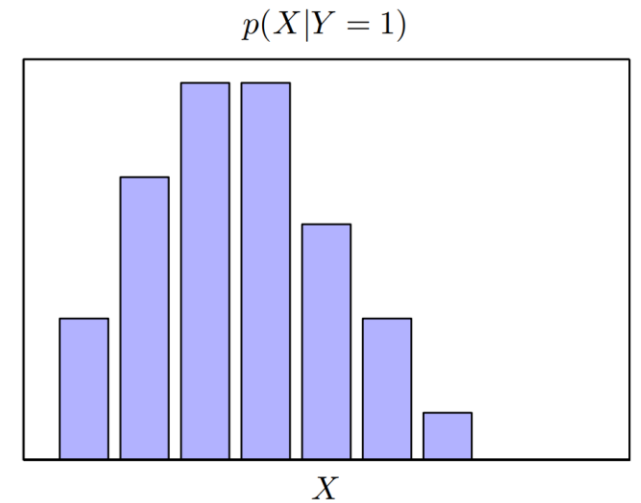
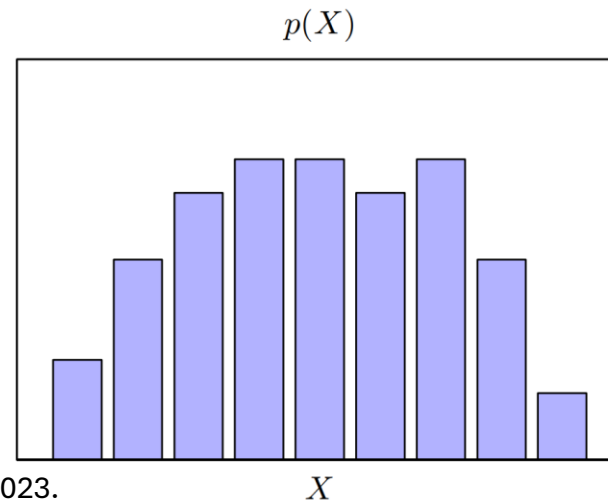
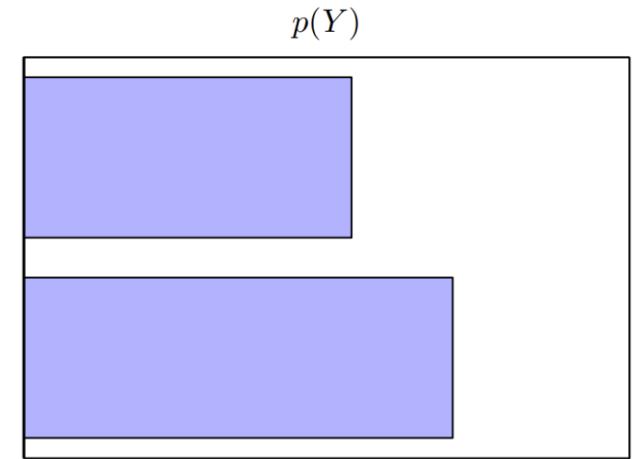
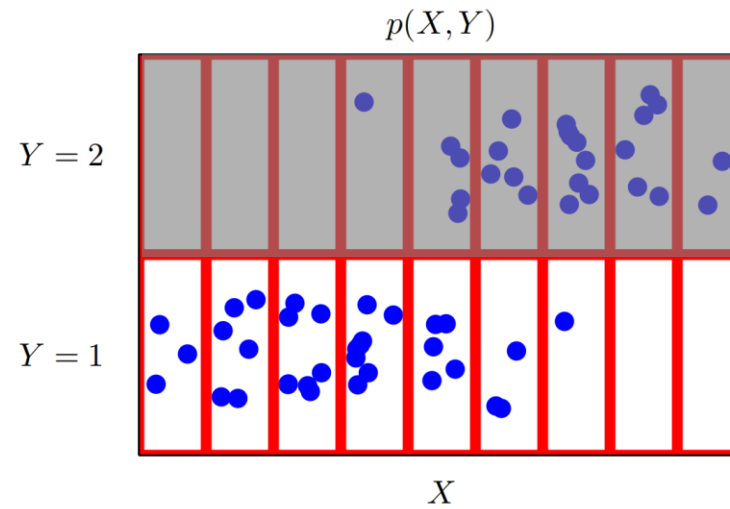
Autoencoders



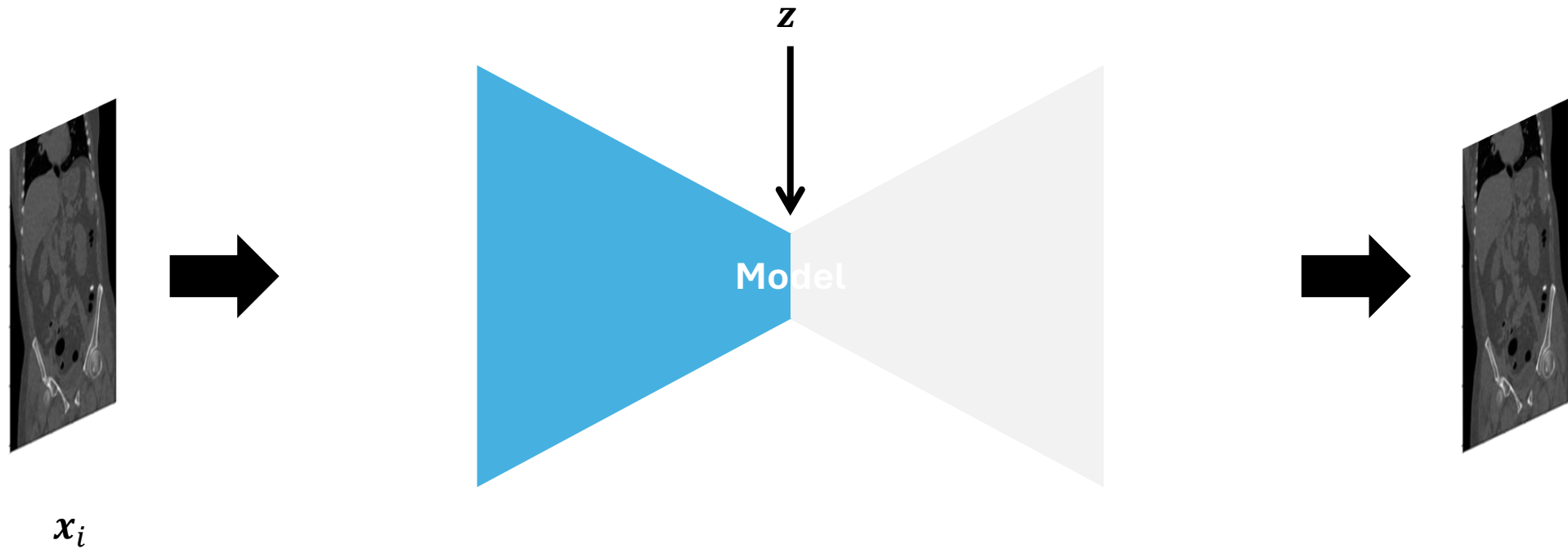
$$p(\mathbf{x}|\mathbf{w}) = \int p(\mathbf{x}|\mathbf{z}, \mathbf{w})p(\mathbf{z})d\mathbf{z}$$

Visual Aid

$$p(Y) = \sum_X p(Y, X)$$



Autoencoders



$$p(\mathbf{z}|\mathbf{x}, \mathbf{w}) = \frac{p(\mathbf{x}|\mathbf{z}, \mathbf{w})p(\mathbf{z})}{p(\mathbf{x}|\mathbf{w})}$$

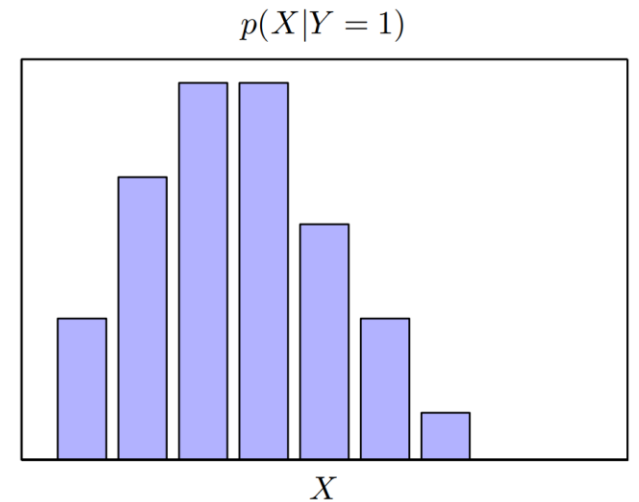
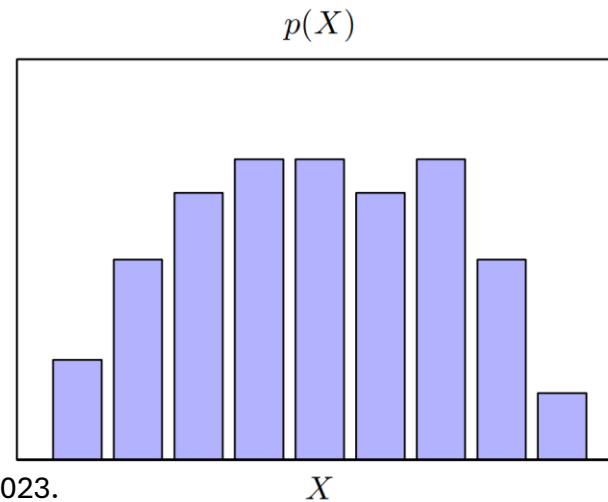
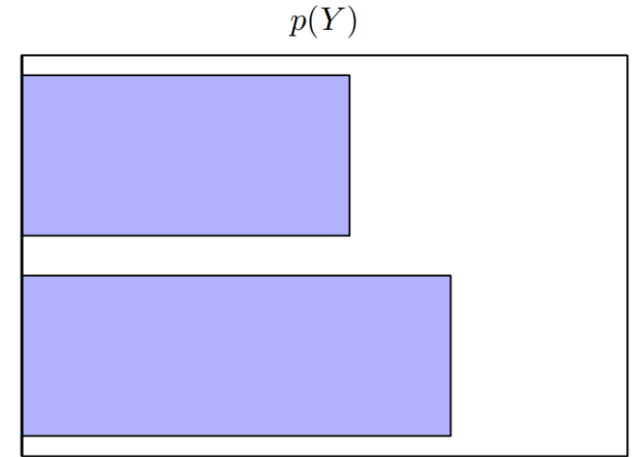
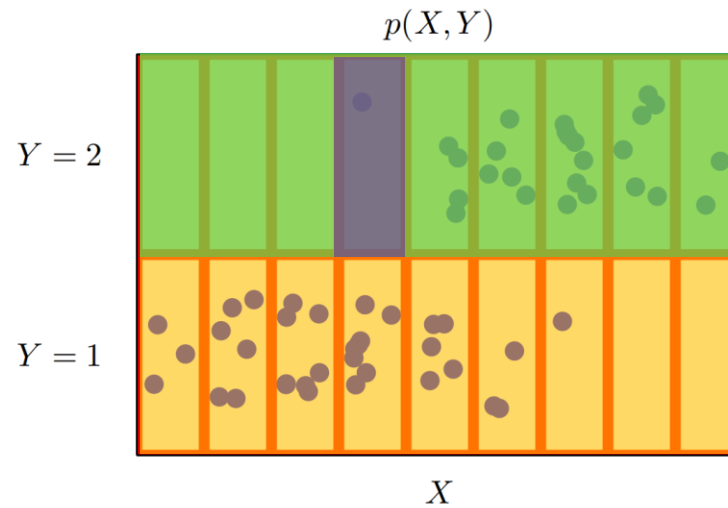
Visual Aid

$$p(X, Y) = p(X|Y)p(Y)$$

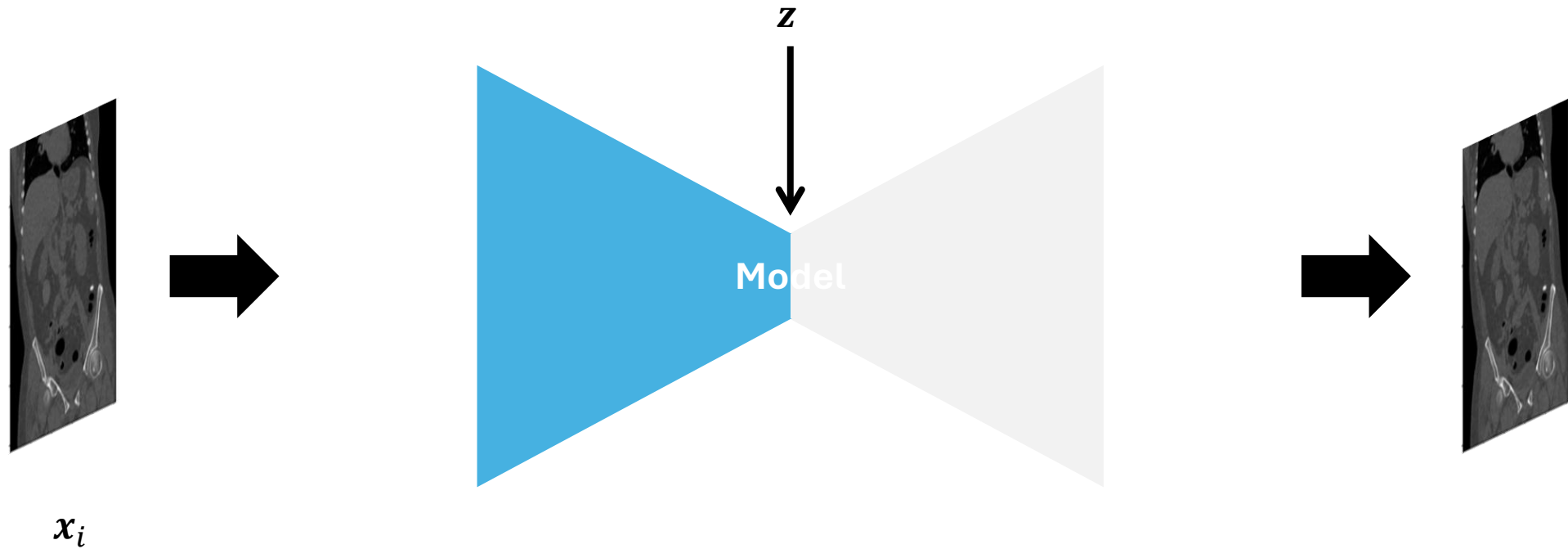
$$p(X, Y) = \frac{n_{ij}}{N} = \frac{n_{ij}}{r_j} \frac{r_j}{N}$$

$$p(Y, X) = p(X, Y)$$

$$p(Y|X)p(X) = p(X|Y)p(Y)$$



Autoencoders



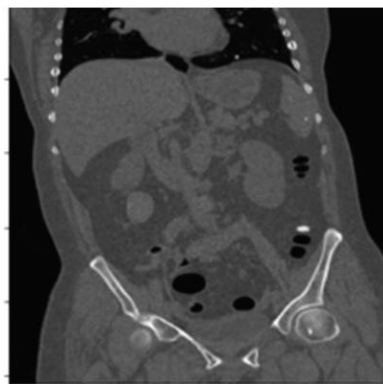
$$p(\mathbf{z}|\mathbf{x}, \mathbf{w}) = \frac{p(\mathbf{x}|\mathbf{z}, \mathbf{w})p(\mathbf{z})}{p(\mathbf{x}|\mathbf{w})}$$

$$p(\mathbf{x}|\mathbf{w}) = \int p(\mathbf{x}|\mathbf{z}, \mathbf{w})p(\mathbf{z})d\mathbf{z}$$

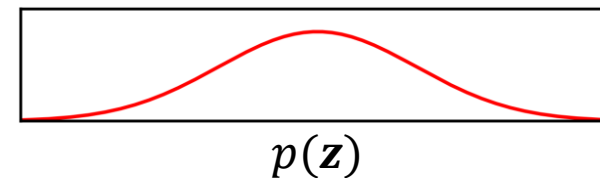
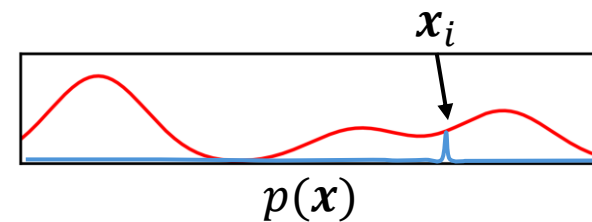
$$p(\mathbf{x}_i|\mathbf{w}) = \int p(\mathbf{x}_i|\mathbf{z}, \mathbf{w})p(\mathbf{z})d\mathbf{z}$$

$$= \int p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} q_i(\mathbf{z}) d\mathbf{z}$$

$$= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right]$$

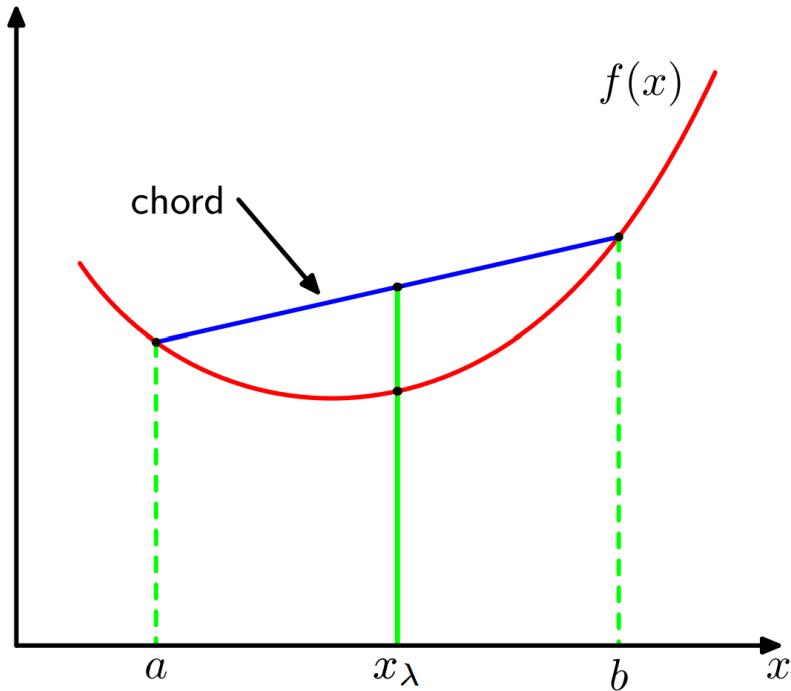


$\mathbf{x}_i$



$p(\mathbf{z}|\mathbf{x}_i, \mathbf{w})$

$$\log p(\mathbf{x}_i|\mathbf{w}) = \log \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right]$$



$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$$

$$\log \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right] \geq \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right]$$

$$\mathcal{L} = \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right]$$

$$\mathcal{L} = \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log q_i(\mathbf{z})]$$

$$\mathcal{L} = \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})] + \mathcal{H}(q_i)$$

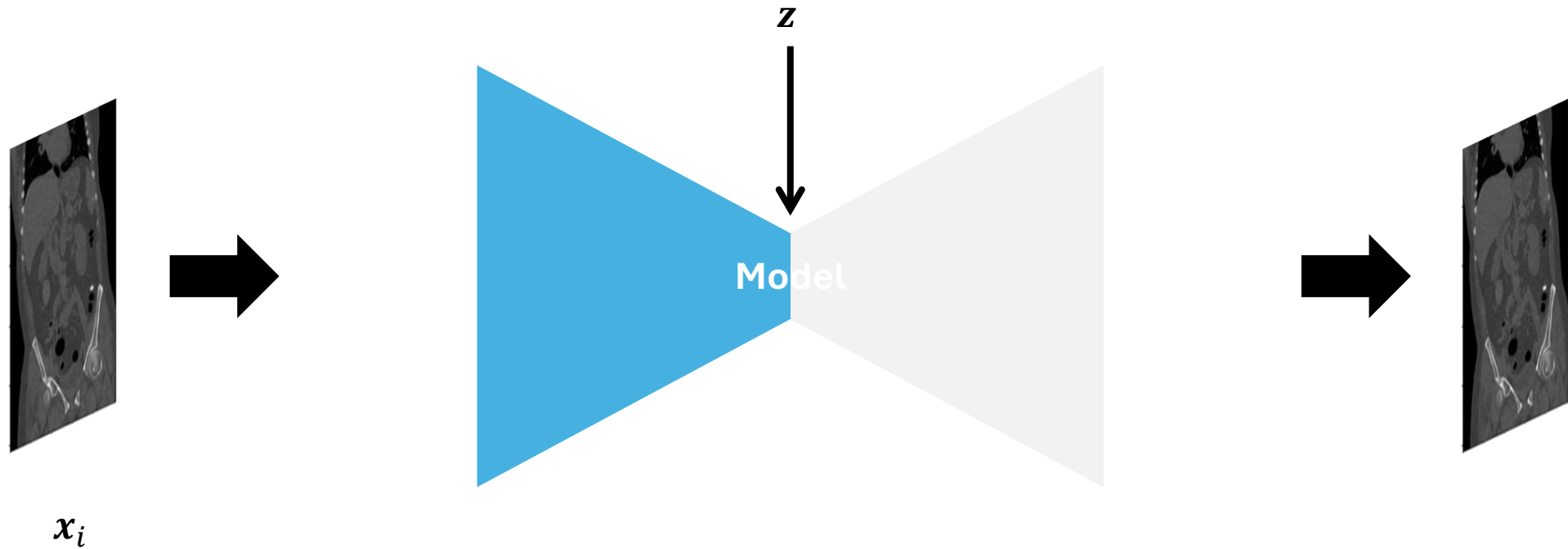


$$\log p(\mathbf{x}_i|\mathbf{w}) = \log \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right] \geq \mathcal{L} = \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})] + \mathcal{H}(q_i)$$

$$q_i(\mathbf{z}) ? \quad p(\mathbf{z}|\mathbf{x}_i, \mathbf{w})$$

$$\begin{aligned} D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}_i)) &= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\log \frac{q_i(\mathbf{z})}{p(\mathbf{z}|\mathbf{x}_i)} \right] \\ &= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log q_i(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{z}|\mathbf{x}_i)] \end{aligned}$$

Autoencoders



$$p(\mathbf{z}|\mathbf{x}, \mathbf{w}) = \frac{p(\mathbf{x}|\mathbf{z}, \mathbf{w})p(\mathbf{z})}{p(\mathbf{x}|\mathbf{w})}$$



$$\log p(\mathbf{x}_i|\mathbf{w}) = \log \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right] \geq \mathcal{L} = \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})] + \mathcal{H}(q_i)$$

$$q_i(\mathbf{z}) ? \quad p(\mathbf{z}|\mathbf{x}_i, \mathbf{w})$$

$$\begin{aligned} D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}_i)) &= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\log \frac{q_i(\mathbf{z})}{p(\mathbf{z}|\mathbf{x}_i)} \right] \\ &= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log q_i(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{z}|\mathbf{x}_i)] \\ &= -\mathcal{H}(q_i) - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\log \frac{p(\mathbf{x}_i|\mathbf{z}, \mathbf{w})p(\mathbf{z})}{p(\mathbf{x}_i|\mathbf{w})} \right] \\ &= -\mathcal{H}(q_i) - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})] + \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{w})] \end{aligned}$$



$$\underbrace{\log p(\mathbf{x}_i|\mathbf{w})}_{q_i(\mathbf{z})?} = \log \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) \frac{p(\mathbf{z})}{q_i(\mathbf{z})} \right] \geq \mathcal{L} = \underbrace{\mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})]}_{p(\mathbf{z}|\mathbf{x}_i, \mathbf{w})} + \mathcal{H}(q_i)$$

$$q_i(\mathbf{z})? \quad p(\mathbf{z}|\mathbf{x}_i, \mathbf{w})$$

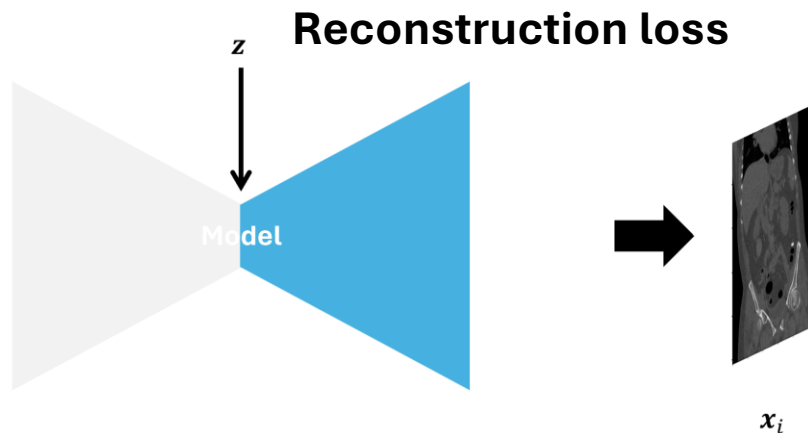
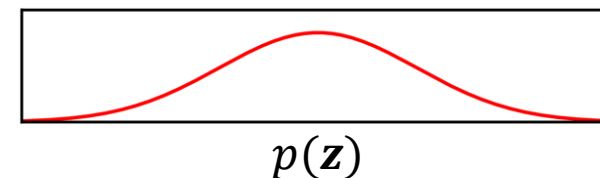
$$D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}_i)) =$$

$$\underbrace{-\mathcal{H}(q_i) - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})]}_{p(\mathbf{z}|\mathbf{x}_i, \mathbf{w})} + \underbrace{\mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i|\mathbf{w})]}_{q_i(\mathbf{z})?}$$

$$D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}_i)) = -\mathcal{L} + \log p(\mathbf{x}_i|\mathbf{w})$$

$$\log p(\mathbf{x}_i|\mathbf{w}) = \mathcal{L} + D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}_i))$$

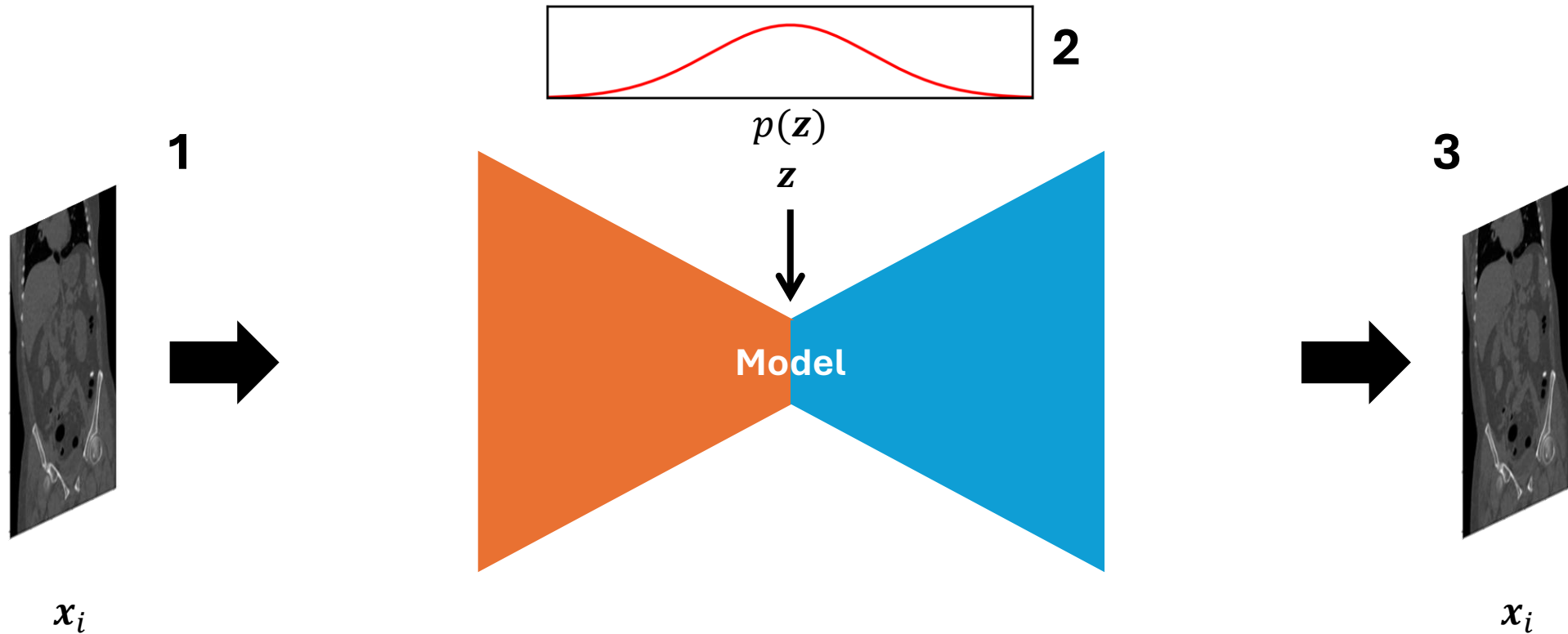
$$\begin{aligned}
\mathcal{L} &= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) + \log p(\mathbf{z})] + \mathcal{H}(q_i) \\
&= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w})] + \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log q_i(\mathbf{z})] \\
&= \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w})] - \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\frac{\log q_i(\mathbf{z})}{\log p(\mathbf{z})} \right] \\
&= \underbrace{\mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} [\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w})]}_{\text{Reconstruction loss}} - \underbrace{D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z}))}_{\text{KL loss}}
\end{aligned}$$



$$D_{KL}(q_i(\mathbf{z}) || p(\mathbf{z} | \mathbf{x}_i)) = \mathbb{E}_{\mathbf{z} \sim q_i(\mathbf{z})} \left[\log \frac{q_i(\mathbf{z})}{p(\mathbf{z} | \mathbf{x}_i)} \right]$$

Variational Autoencoders

*D. Kingma and M. Welling, arXiv:1312.6114. 2013. **D. J. Rezende, S. Mohamed and D. Wierstra, ICML. 2014.

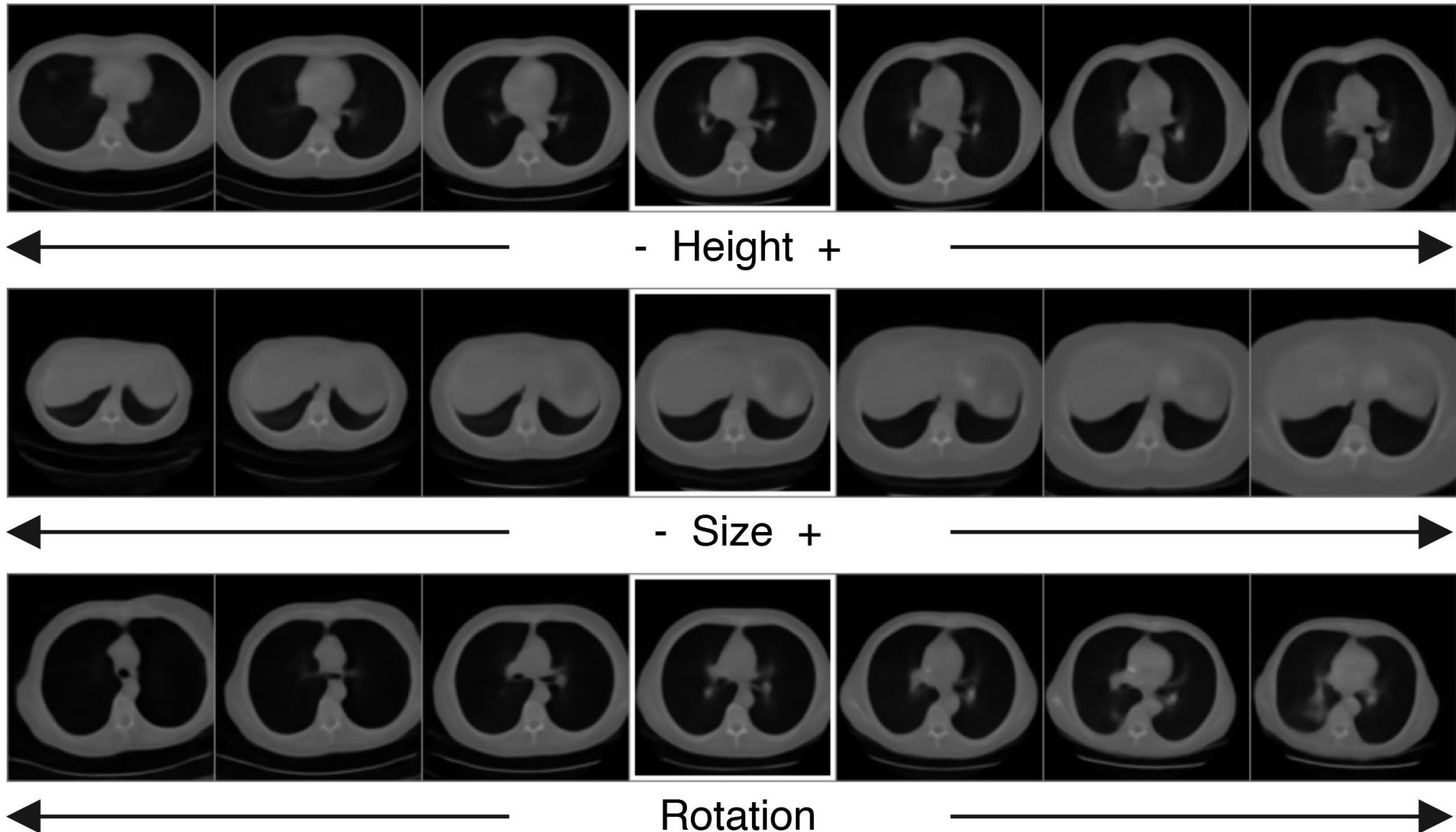


$$\mathcal{L} = \mathbb{E}_{z \sim q_i(z)} [\log p(x_i | z, w)] - D_{KL}(q_i(z) || p(z))$$
$$\mathcal{L} = \underbrace{\mathbb{E}_{z \sim q(z|x, \phi)} [\log p(x|z, w)]}_{\text{Reconstruction loss}} - \underbrace{D_{KL}(q(z|x, \phi) || p(z))}_{\text{KL loss}}$$

Latent Space Interpolation



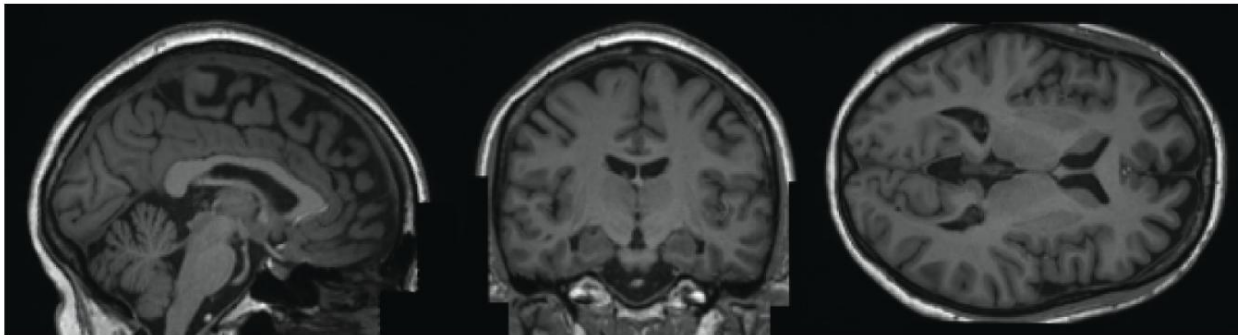
Medical Latent Interpolation



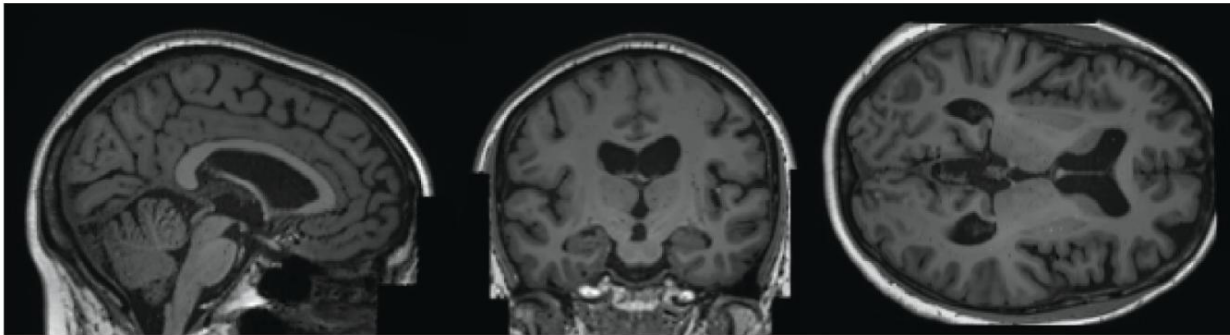
Better Generative Models

Sample 1

Real



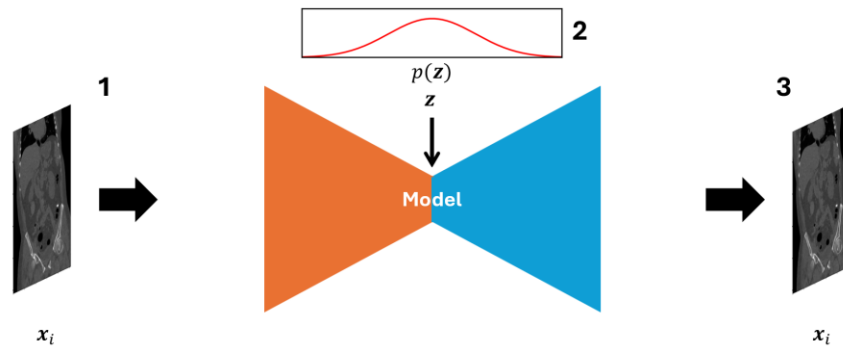
LDM



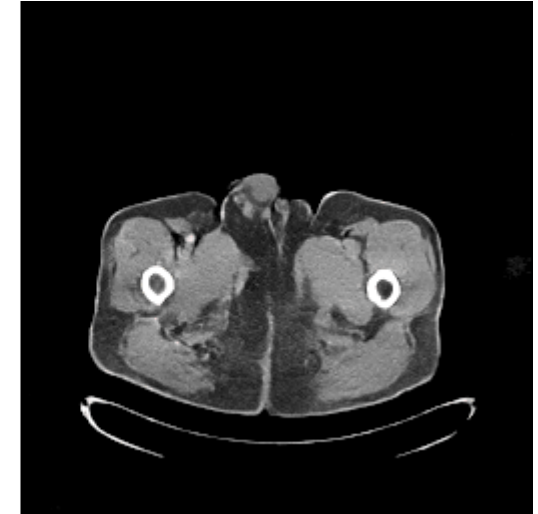
MAISI-v2



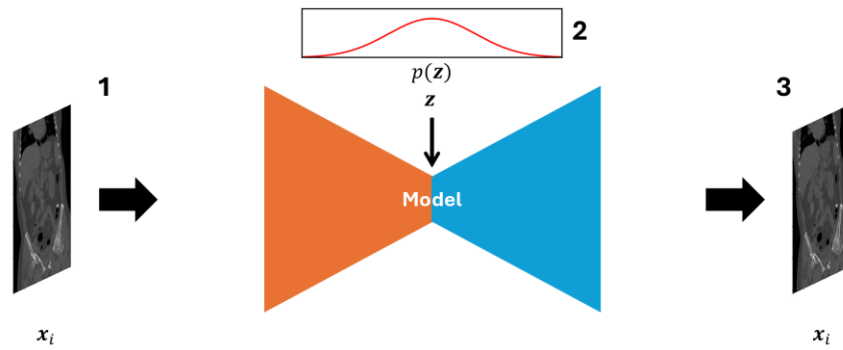
Now What?



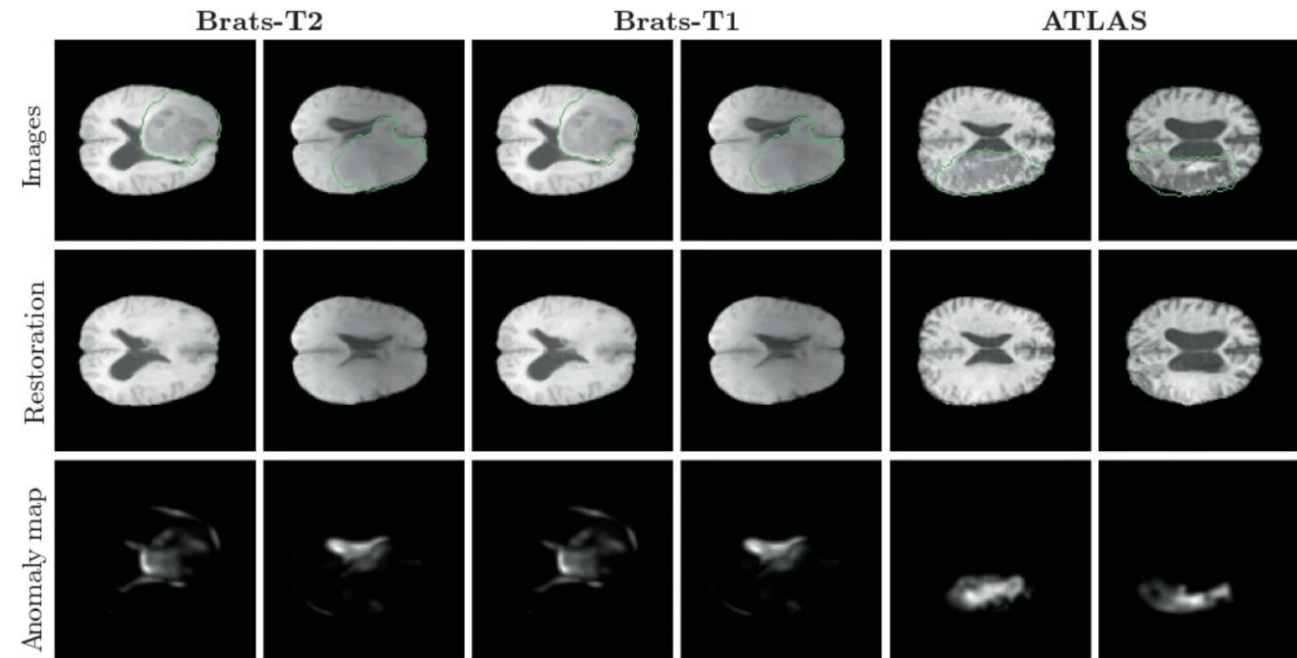
Generative Model



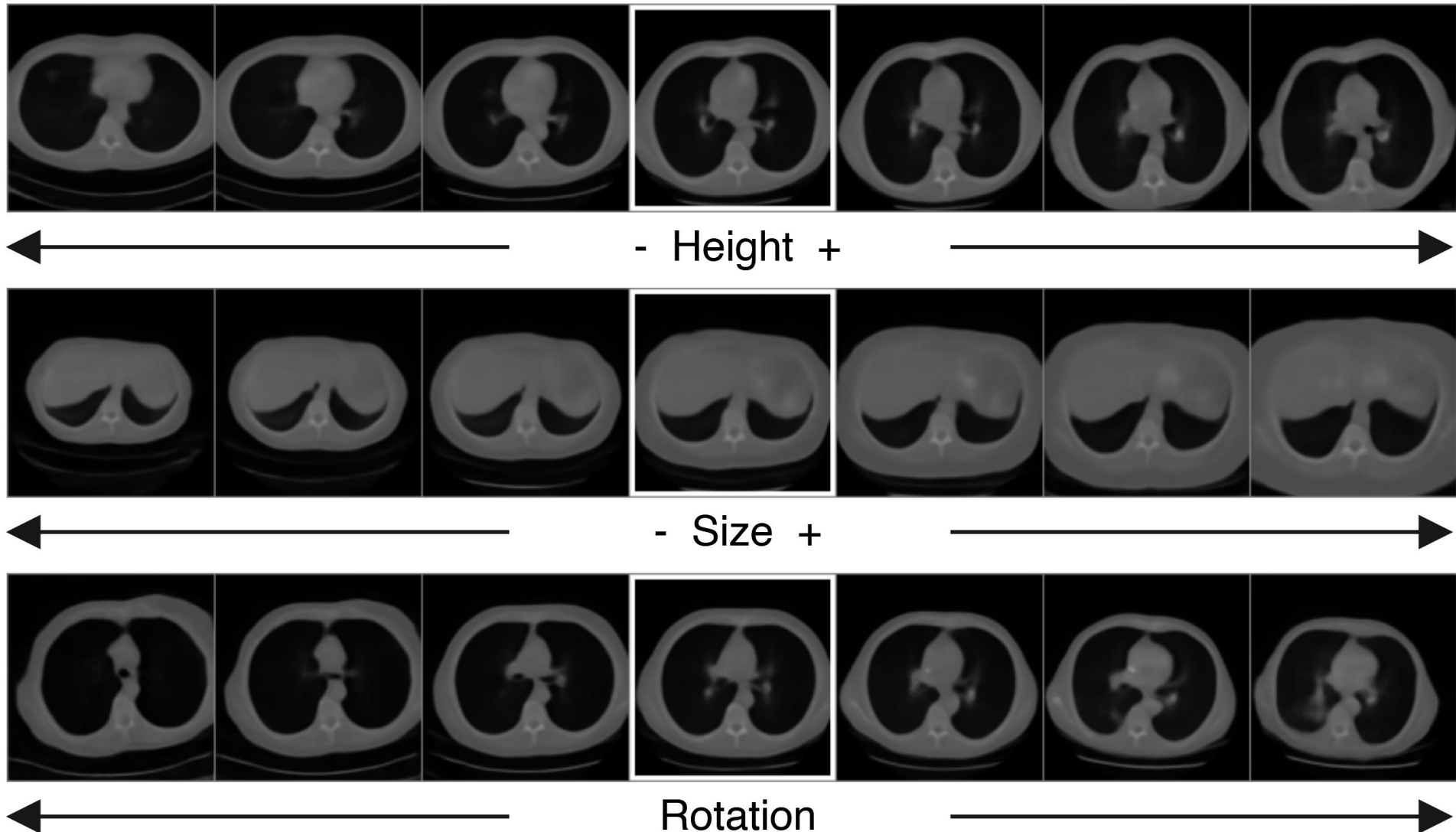
Now What?



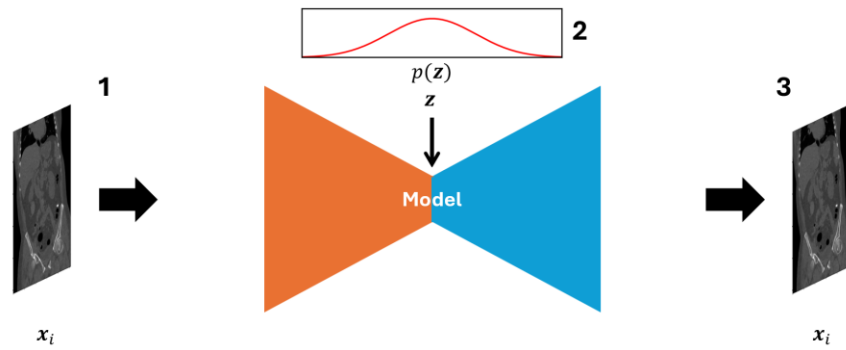
Generative Model



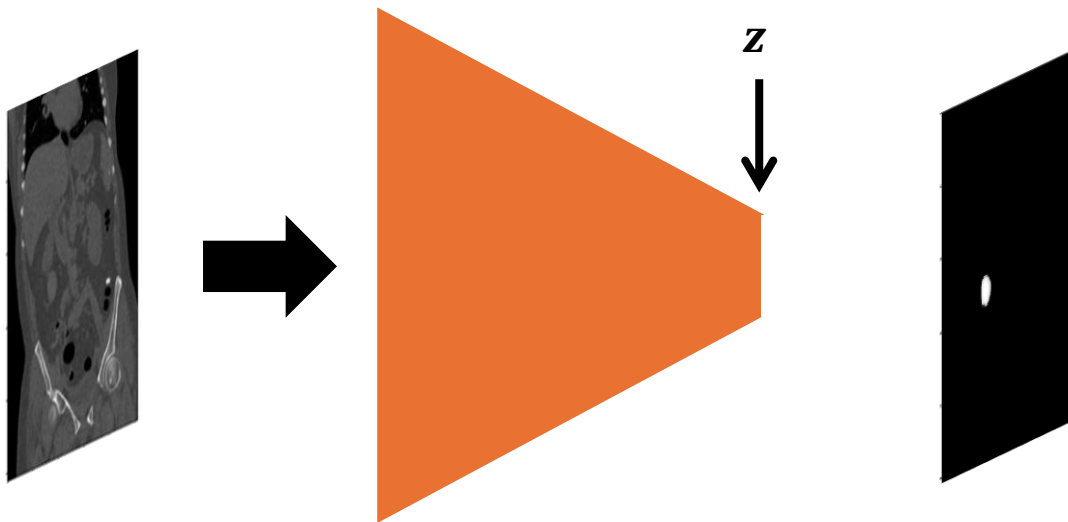
Medical Latent Interpolation



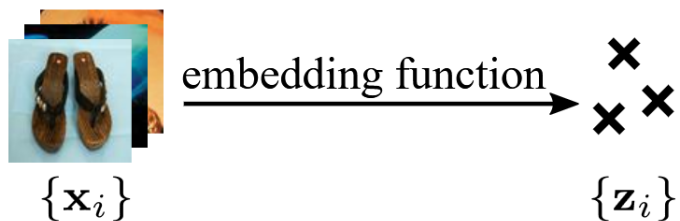
Now What?



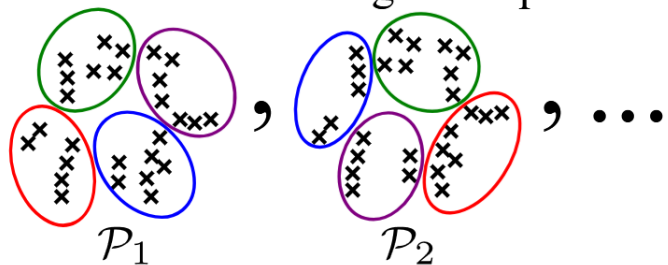
Clustering/Classification/Segmentation



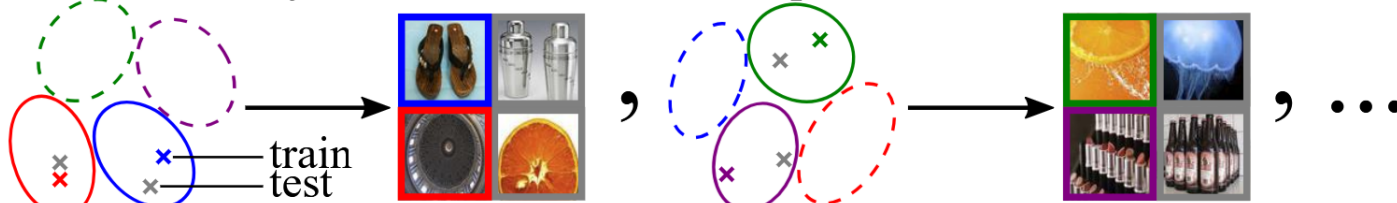
1. run embedding learning



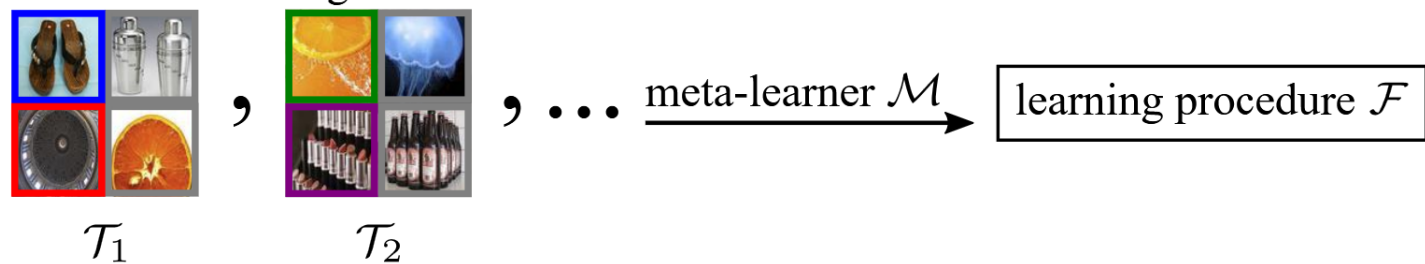
2a. cluster embeddings multiple times



2b. automatically construct tasks without supervision



3. run meta-learning on tasks



miniImageNet



CelebA



What we have?

- The data itself (images)
- Domain knowledge of:
 - The sampled distribution
 - The subject
- Assumptions

Vs.

What we want?

- Approximate features of supervised model
- General features for any task
- Manifold where data lies
- Cluster/biomarker discovery

Tutorial #1

AI 4 Life Science - Unsupervised Representation Learning - Tutorial 1: Introduction and VAEs

Setup and Groups

Form groups of 3, ideally with people you do not work with regularly. Take 5 min each to explain a bit about your work:

1. What is the focus of your work?
2. What is the limitation or most difficult technical aspect?
3. Could unsupervised learning fit into your project in any way? If so, where could it improve things. If not, why is it unhelpful and what do you wish it could do?



https://github.com/jemt看/Al4LS_SummerSchool_UnsupervisedRepresentationLearning